Simulating a Labor Time Economy

An Overview

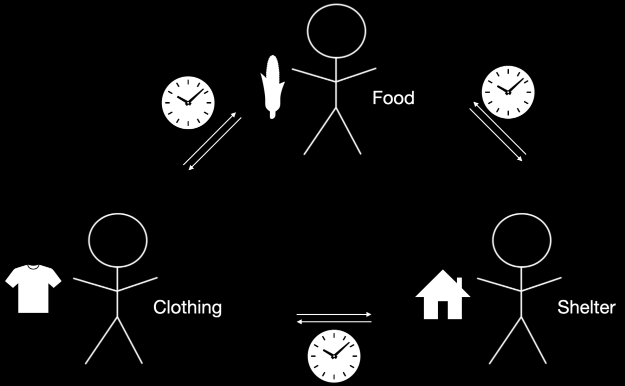
A. Aviram, A. Creiner, D. Lim, N. Lim

August 2026

1. Introduction

What are Labor-Time Credits?

- Value comes from work —– time taken from life to make or do something that someone needs.
- Labor-time credits, like money, make goods and services commensurable.
- But they do not represent profit or rent and do not circulate freely.
- They are a *fiat credit* currency: issuance creates debt, a promise to produce something that someone will consume.



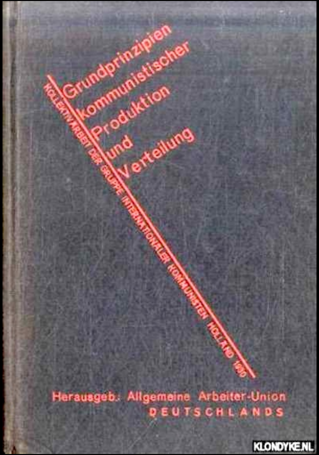
What Is a Labor-Time Economy (LTE)?

A labor-time economy (LTE) is a socialist economy in which

- Private ownership of the means of production is abolished — "property" in the common modern sense has no meaning.
- People work individually or in worker-managed collectives (firms) to produce and distribute goods and services.
- Workers are paid in labor-time certificates.
- Workers buy what they need in exchange for labor-time certificates.
- Prices correspond to average socially necessary direct and indirect labor time.

The Group of International Communists (GIC)

The German Group of International Communists (GIC) championed an LTE in the 1930s.



What Makes the GIC Model Different?

- The GIC model of an LTE features
  - No central planning of low-level details — planning is democratic and mostly decentralized.
  - No five-year plans or other fixed production periods — producers respond autonomously to consumption demands.
- These features encourage the *free association of producers* and are intended to avoid
  - Top-down bureaucratic control
  - Worker alienation
  - Exploitation
  - Replication of capitalist-style employer-employee relations of production.

Why Choose a Labor Time Economy?

- Presupposes worker ownership of the means of production.
- Prevents exploitation — i.e., extraction of surplus value.
- Prevents alienation — from product, labor, self, and others.
- Bars currency circulation, hoarding (credits are used up like theater tickets), asset accumulation, rents, and rent seeking.
- Maintains the tautological equivalence between labor performed and total output price.
- Ensures that workers can always afford their consumption.
- Rewards ingenuity, innovation, and cooperation: less work, higher living standards, or both.

Why Choose a Labor Time Economy?

As in a money economy:

Accounting $\rightarrow$
Signals $\rightarrow$
Decentralized Planning + Responsiveness $\rightarrow$
Autonomy + Democracy

- Buying and selling seem natural to people already used to money exchanges — BUT
- The LTE avoid distortions made possible by money — profit, market manipulation, monopoly rents, inflationary and deflationary cycles, etc.
- Local rationality sustains global stability.
- LT credits keep the useful aspects of money while eliminating the harmful ones.
- The LTE provides a *pragmatic transition* out of capitalism.

What Does it Mean to Choose a LTE?

- The transition to a LTE involves a revolution in thought:
  - Prices represent the total socially necessary labor time needed to produce things.
  - "Money in the bank" is not an asset to be hoarded or traded. Credits are a reserve of social entitlement, to be redeemed for a portion of the social product.
  - Society must maintain accurate estimates of these socially necessary labor values and transparent information on how labor is distributed for production.
  - Society commits to full employment for all who can work.
  - Society commits to the equal distribution of the benefits of its labor — sustenance, free time, human flourishing.
- With these choices made, along with the abolition of private property, an LTE similar to what we are about to present would likely emerge with or without our guidance.

Advantages We Have Demonstrated So Far

- Our simulated LTE *computes* labor values as a *decentralized*, *distributed* system — starting from arbitrary initial prices and *converging* to real labor values in linear time.
- Individuals and firms making reasonable local decisions enable supply to meet demand consistently.
- Taxation needed for a public sector can also be computed, adjusted, and applied automatically.
- Thus governance of essential, basic economic operations as a distributed automaton *frees up the political sphere* for high-level policy decisions.

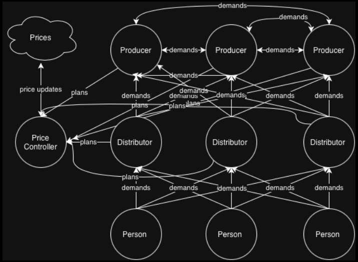
Principles for Simulating the LTE

Our simulation attempts to capture reality in *three layers*:

1. **The System**
   Public information, laws, rules, mechanisms, institutions, and norms essential to the LTE's functioning.
   Encapsulated in data structures, functions, and algorithms.
2. **Reasonable Behavior**
   Our best guess at what an individual or firm would do in response to available information.
   Modeled in algorithms — e.g., reordering to replenish firms' input inventories.
3. **Unpredictable Behavior**
   Individual or worker co-op decisions, unforeseen events (e.g., illness), personal characteristics (e.g., skills, predilections, or tastes).
   Modeled with random or pseudo-random values.

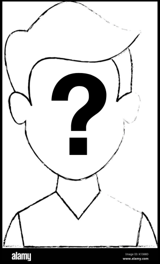
Three Kinds of Simulation

Our simulated world includes three categories:



1. System

2. Reasonable Behavior



3. Unpredictable Choices

System Information

- Information available to everybody:
  - Prices
    - Of goods — reflecting the total average cost of production.
    - Of consumer goods — including the cost of distribution.
  - Standard work day and week.
  - Industry standard production methods, including
    - Quantities of inputs
    - Machines needed.
  - Each firm's production plan history and account record.
  - Each firm's *catalog* of goods produced or distributed.
- Information available to each producer and distributor:
  - Purchases from downstream, serving as signals of demand.
  - Each worker's skill level.

Simplifying Assumptions in Our Simulation Model — For Now

- The economy is autarkic and self-sufficient.
- Each good comes in only one kind — with no competing "brands."
- Services are not modeled as distinct from goods.
- Each good has only one (standard) production method, with known input and machinery requirements, and each production methods output only the respective goods.
- Labor is homogeneous.
- A machine is worn down *through use*, but at a *constant rate*.
- Machines stand in for all sorts of fixed capital.
- The only factors that affect an individual's pace of work are skill level and health status.
- Worker management time is simply factored in as part of production.
- The population is fixed and stable.

2. Overview

Basic Overview

- Our model is discrete and **agent-based**: agents of various kinds interact through a fixed set of rules (protocols).
- On each *time step*, all agents are called on to perform a set of actions unique to that type of agent. These actions include consuming, drafting plans, shopping, and so on. (The model is implemented serially, for now, so agents act one at a time.)
- We assume conceptually that a single round of all agents executing their set of actions constitutes one hour of time passing within their "world".
- Our model has two types of agent: **persons** and **firms**.
- Firms come in two types: **producers** and **distributors**.
- Our overview will present what a time step looks like for each of these different kinds of agent.
- Before that though, let us summarize some of our guiding design principles.

Grounding Design Principles and Goals

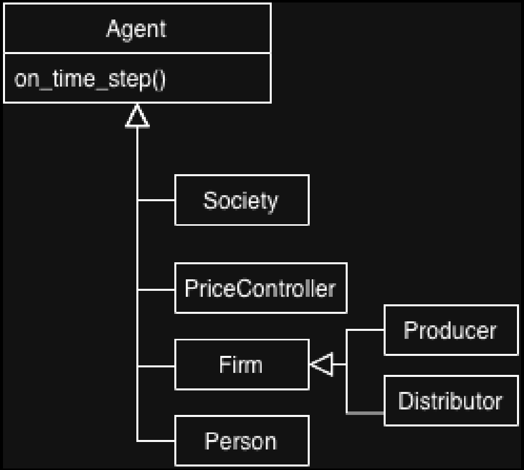
- Assume that firms do their best to satisfy orders placed with them, use their best judgment when placing orders, place their orders in a timely manner and with reasonable expectations, and show solidarity to one another as well as individuals.
- Build our algorithms as realizations of what the coordination of these firms towards economic homeostasis *would* look like. This is to be contrasted with a more prescriptive approach (e.g. what *should* be done).
- Wherever possible, work to ensure that the decisions being made by the agents *make sense*, given the above behavioral assumptions. We call this *local rationality*.
- Wherever possible, design systems that *align the incentives* of all agents.
- *Only where not possible*, design negative feedback systems which patch up any disalignment of interests. So far, we have not done this at all, but we have identified several points where it is likely necessary.

Grounding Design Principles and Goals

- Keep jargon to a minimum and keep solutions understandable.
- Determine a minimal set of protocols and communication channels needed to achieve economic homeostasis.
- Make sure that all necessary information for system operation is easy to obtain from within the system itself (without requiring complex or difficult external data analysis).
- Determine the limits of what can be accomplished *without* any kind of central planning.
- An economic social organism, as a cognitive agent, has a cerebellum (cybernetic Systems II – III) as well as a cerebrum (Systems IV – V).
- The exercise we are focused on is determining what aspects of the economy can be left to the cerebellum. Only after this can we properly understand the role of the cerebrum (centralized democratic planning).

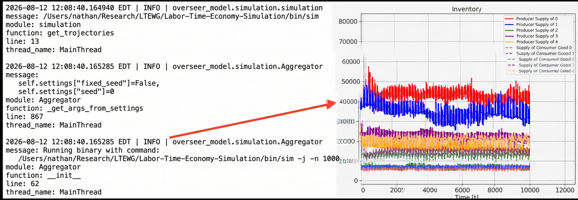
System Architecture: Agents & The Simulation Loop

- We translated our economic rules into a discrete, object-oriented C++ simulation.
- The system is driven by an **Agent** base class.
- Every active entity (Persons, Firms, Price Controller) overrides a core `on_time_step()` function.
- During each simulated hour, the engine calls this function for all agents, triggering a **cascade of simulated events** (e.g., shopping, drafting plans) unique to that subclass.



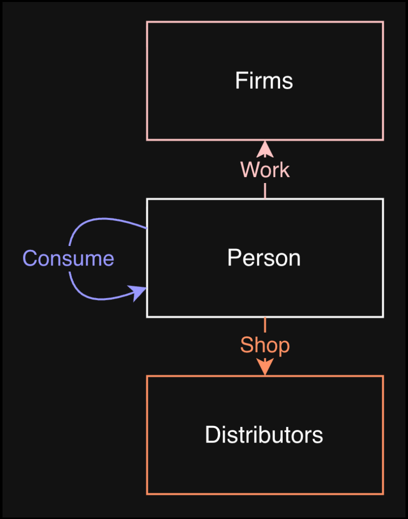
System Architecture: Overseer Data Flow

- As the `on_time_step()` loop runs, various routines continuously export data of interest in JSON format.
- These data include shifting firm inventories, real-time price fluctuations, and persons' labor credits.
- The Overseer tool ingests this output to generate visual graphs and track the economic homeostasis of the system over time.



Basic Overview: Persons

- They possess an inventory of goods as well as an account with a number of labor credits, which they can spend on goods and earn through laboring at a firm which employs them.
- Each time step, a person reasonably decides to do the following:
  1. Consume from their inventory according to consumption frequencies.
  2. Shop, if their inventory is in enough deficit. Their goal is to maintain a stockpile that will last them a certain amount of time. Think: "I only want to have to go shopping once a week, while still always having what I need."
- They are also impacted by the following outside their control:
  1. Possibly getting sick if healthy, or recovering if sick.
  2. Assignment to production plans within a firm.

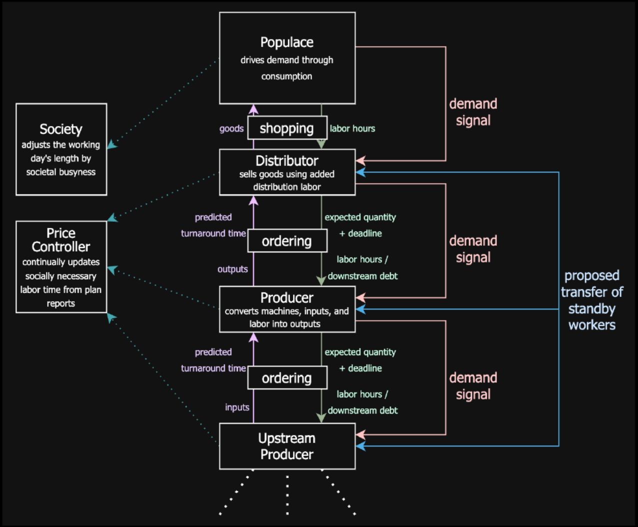


Basic Overview: Distributors

- The job of a distributor firm is to make consumer goods available to persons for purchasing.
- The act of taking an ordinary good and turning it into a consumer good requires labor (stocking shelves, transportation). Thus they have employees just like regular firms.
- As a distributor makes sales, its supply of consumer goods dwindles. As this happens, they place orders with themselves to "restock their shelves".
- Eventually, their supply of "unshelved" goods dwindles, at which point they begin placing restocking orders from producers. (More details of this will follow when we cover firms generally.) This stimulates economic activity from the producers.

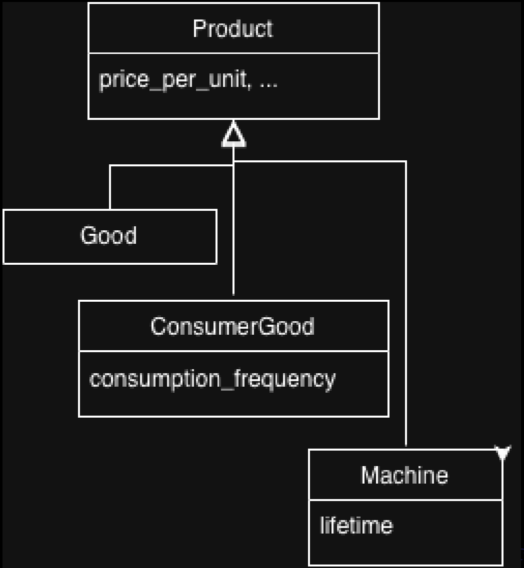
Basic Overview: Economic Metabolism

- As the producers' own stockpiles of required inputs dwindle, they begin placing orders with one another to replace these capital goods.
- In this way, consumption from persons is the sole driver of production. Eventually, this leads to a propagation of demand through the entire supply chain.



Basic Overview: Products

- There are three types of **product** within our simulation: **goods**, **consumer goods**, and **machines**.
- Each good has a corresponding consumer good, and each consumer good requires exactly one unit of the corresponding good plus some living labor to distribute it.
- As mentioned, this represents the act of transporting and stocking the good on the shelf for a consumer to purchase.
- Machines represent fixed capital. Each machine, in addition to possessing the normal properties of goods, possesses a lifespan in hours, which gets depleted as the machine is used in the production process.



Basic Overview: Plans and the Price Controller

- When a firm needs something, it create orders, and sends those orders out to producers of that thing. These firms draft plans to satisfy the order.
- A **plan** is a fundamental unit of representation of production, represented as data, which is dynamically updated as its associated production process progresses.
- When an order is completed, the completed plan is passed on to a price controller algorithm, which uses the data to update its prices.
- Through the dynamic action of the price controller, an identity is established between the labor done and the labor credits spent, ensuring not just that workers are never left lacking the credits they need to purchase what they want, but also the fairness of labor compensation.

Initialization Overview

- Our system is defined by many parameters (too many!). Some important ones are shown on the right.
- We employ a classical Leontief/Sraffa/Morishima linear production model. This means:
    - Fixed techniques of production, one per type of good.
    - Each technique produces exactly one unit of its respective good, with no byproducts.
    - Linear returns to scale.
- We plan to extend our system to a more robust joint production model in the near future.

Parameter	Meaning
n_p	Num. persons
n_g	Num. goods
n_m	Num. machines
$n = 2n_g + n_m$	Num. products
n_r	Num. producers
n_d	Num. distributors
T	Hours / working day
W	Days / working week

Initialization Overview

- Index the goods $1 \dots n$: normal goods first, then machines, then consumer goods. Identify goods by their index.
- Under the Leontief assumptions, each good type i has a vector bundle of requirements for the production technique which creates it $\mathbf{a}_i$. Additionally we have a living labor vector $\mathbf{l}$.
- a_{ij} represents the number of units of good i required to create a unit of good j, and l_i represents the number of hours of direct labor from a single worker are required to produce good i.
- The living labor values are chosen randomly at initialization, as are the numbers a_{ij} where i is the index of a good (i.e. for $i = 1, \dots, n_g$).

$$\mathbf{a}_i = \begin{pmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{pmatrix} \tag{1}$$

$$\mathbf{l} = \begin{pmatrix} l_1 \\ l_2 \\ \vdots \\ l_n \end{pmatrix} \tag{2}$$

Initialization: Production Techniques

- For machine m with lifetime $\mathcal{L}$, to determine the value a_{mi} for some i, we first estimate the average team size A as the number of persons divided by the number of firms. If l_i is the living labor required to produce a unit of i, then the proportional deduction from the lifespan of the machine is

$$a_{mi} = \frac{l_i}{A\mathcal{L}} \tag{3}$$

- With all that what we've said, the overall Leontief input/output matrix has the block form

$$A = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n \end{pmatrix} = \begin{pmatrix} A_{gg} & A_{gm} & I_c \\ A_{mg} & A_{mm} & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{4}$$

where I_c is the $n_g \times n_g$ identity matrix, A_{gg} is the matrix of goods requirements for goods, and A_{gm} is the matrix of goods requirements for machines, and so on.

Initialization: Production Techniques

- Once we have initial values of all entries of this matrix, we must normalize it to ensure productivity (Hawkins-Simon condition) of the submatrix

$$A_p = \begin{pmatrix} A_{gg} & A_{gm} \\ A_{mg} & A_{mm} \end{pmatrix} \quad (5)$$

- This is to ensure that social reproduction is, at the very least, possible. Additionally, it ensures that gross bundles as well as labor values are always well defined.
- To do this, we multiply the entries of A_p by the scalar $\frac{\epsilon_A}{\rho(A_p)}$, where $\rho(A_P)$ is the spectral radius of A_p, and $\epsilon_A < 1$ is a parameter of the system that we call the **difficulty of production**. The closer it gets to one, the more difficult it is for the society to produce a surplus.
- Finally, the lifetimes of the machines must be adjusted according to this inverse of this scalar, since they are implicitly altered by our normalization. These normalized coefficients are taken as defining the production technique requirements.

Initialization: Abilities

- Labor is considered homogeneous in the classical Leontief sense. However, not all labor is considered *equal* in our simulation.
- Each good has a set of skills associated with it, and workers have varying levels of proficiency at these skills, which act as modifiers to the living labor requirements.
- The number of skills is controlled with a parameter n_a. Skills are assigned randomly to the goods. The number of skills assigned per good is a fixed parameter, and skills are chosen uniformly at random without replacement.
- For each person, proficiency levels are assigned in a lognormally distributed manner with mean 1.
- For a single worker laboring to produce good i, requiring the singular skill which the worker has the proficiency level s in, then the living labor which will be required is sl_i.
- For a good requiring multiple skills, the modifier s instead becomes the average of each relevant proficiency.

Initialization: Sickness

- People in the economy will occasionally become ill, taking time to recover. While ill, they can continue to work, but their labor proficiency is cut in half.
- The annual proportion of society which will fall ill annually is a constant parameter of the model, which we set to 0.1 by default - the average percentage of the United States which contracts the flu each year.
- Workers have a chance to recover each time step. The average days to recovery is to 5 - the average duration of flu symptoms.
- These two less conventional factors - proficiency and illness, allow us to witness more interesting dynamics from the price controller and test its robustness.
- More specifically, the socially necessary labor times are no longer strictly equal to those predicted by the techniques of production. They are an unpredictable, emergent social average.

Initialization: Consumption

- While the condition of productivity is well known to economists as a necessary condition for any realistic economy, the conditions of *consumption* are less well understood.
- Let $\mathbf{c}$ denote the vector representing the average **hourly consumption frequencies** for a single person.
- Clearly, some limitations must be placed on $\mathbf{c}$ in order to ensure that society is not consuming faster than it can produce.
- In the context of studying capitalism, these conditions are typically smuggled in (perhaps accidentally) through the standard profitability assumptions, which place indirect bounds on the real wage.
- In studies of post-capitalist economies, consumption bounds must still be imposed. What form do these conditions take when liberated from the profit equations?

Initialization: The Reproduction Condition

- The maximum number of hours that a single person can work *per hour* is given by

$$\frac{TW}{7 \times 24} = \frac{TW}{\Gamma} := \omega \tag{6}$$

- Simply put, for social reproduction to be possible, we must have it be the case that the average amount of total labor value that a single person consumes per hour does not exceed the maximum amount of labor value that a single worker can produce per hour. This maximum amount of producible labor value per hour is the quantity above: ω.
- Let $\mathbf{v}$ denote the vector of unit labor values. Mathematically, the condition can be expressed as

$$\mathbf{v} \cdot \mathbf{c} \leq \omega \tag{7}$$

- The number $\frac{\mathbf{v} \cdot \mathbf{c}}{\omega}$ in our model is analogous to what Cockshott et. al. refer to as the **reproduction coefficient** in one of their models constructed in *Classical Econophysics*. It is the kernel of socially necessary truth buried within the standard profitability assumptions of capitalist economics.

Initialization: Consumption

- To ensure this condition, we draw a random set of numbers to act as our initial consumption frequencies for normalization. We then apply the normalization

$$\mathbf{c} \mapsto \frac{\omega\epsilon_c}{\mathbf{v}\cdot\mathbf{c}}\mathbf{c} \tag{8}$$

  where $\epsilon_c < 1$ is a constant parameter of the model. Solving for ϵ_c reveals it to be the ratio of the minimum number of hours a person must work in a week divided by the number of hours in the working week.
- This normalization ensures that $\mathbf{v}\cdot\mathbf{c} = \omega\epsilon_c < \omega$.

Initialization: Wrap Up

- This summarizes the details of initialization which are interesting enough to warrant mention. To briefly summarize other relevant details:
  - With $\mathbf{c}$ defined, the net hourly demand bundle is predictable: $\mathbf{y} = n_p\mathbf{c}$. Firm-local demand projections are drawn randomly from a normally distribution with means set according to the gross hourly demands $\mathbf{x} = (I - A)^{-1}\mathbf{y}$.
  - Firm inventories are then initialized by scaling up these numbers to give them a stockpile which is roughly sufficient for a certain amount of operating time.
  - Similarly, persons are initialized with enough labor credits in their accounts to make their normal purchases for a fixed amount of time according to the consumption frequencies vector $\mathbf{c}$. Additionally they are given a week's worth of goods in their starting personal inventories to consume from.
  - Prices can be initialized in one of two ways: At their labor values as predicted from the requirements matrix and living labor vector, and at their Perron-Frobenius equilibrium prices of producton.[1]

[1] This requires an additional real wage vector, which we obtain by repeatedly scaling down the consumption frequency vector $\mathbf{c}$ until a certain level of profitability is assured.

Operation: Firms

- All firms maintain a **catalog** C listing goods that they produce. These catalogs are in principle publicly available to all other firms for the sake of placing orders.
- Each time step, a firm will do the following:
  (1) For each necessary input to production, place an order for resupply, if necessary, informed by the *demand*.[2]
  (2) Progress any and all active plans.
  (3) Once weekly, allow workers to transfer to other firms if they have a surplus.

[2] Again, groups interested in adopting our model should be encouraged to use their own best judgement when deciding when and how to place orders. Our algorithmic procedures for placing orders are meant to act as a stand-in for that behavior, not as its replacement.

Demand

- By the **demand** for the product, we mean the rate of depletion of a firm's inventory, due to sales, or to usage in a production or distribution process.
- A recursive formula used to estimate demand at time t (D_t) is given by

$$D_t = Q_t \frac{1}{\Delta} + D_{t-1}\left(1 - \frac{1}{\Delta}\right) \tag{9}$$

where Q_t is the amount of the good used during the time step, and Δ is an averaging time constant. Clearly, $D_t = Q_t$ is a steady state solution.

Demand

- For usage distributions which are stable over time ($\exists \bar{Q} \in \mathbb{R}. \forall t \in \mathbb{N}. E(Q_t) = \bar{Q}$), recursive expansion and manipulation shows convergence independent of D_0:

$$\lim_{t\to\infty} D_t = \frac{1}{\Delta}\sum_{i=0}^{\infty}\left(1-\frac{1}{\Delta}\right)^i Q_{n-i} \tag{10}$$

$$\lim_{t\to\infty} E(D_t) = \frac{1}{\Delta}\sum_{i=0}^{\infty}\left(1-\frac{1}{\Delta}\right)^i E(Q_{n-i}) = \frac{E(Q_{n-i})}{\Delta(\frac{1}{\Delta})} = \bar{Q} \tag{11}$$

- Alternatively, for continuous Q and D, an analogous differential equation is

$$\frac{dD}{dt} = \frac{(Q-D)}{\Delta} \tag{12}$$

with the following solution where Δ can be thought of as a *decay rate*

$$D = ke^{-t/\Delta} + Q \tag{13}$$

Inventory and Orders

- Let i be a product the firm is considering placing a reorder for. Reasonably, a firm wants to maintain a *target stockpile* for i, S_i, on which two lower bounds are enforced:

$$S_i = \max \left\{ S_d D_i,\ S_m a_{ji} \ :\ j \in C \right\} \tag{14}$$

where S_d is the *stockpile duration* and S_m is the *stockpile minimum*. The former represents a duration for which production can continue with no resupply (and is generally dominant); the second ensures that the Firm can produce a minimal amount on a full stockpile.

- A logical condition is to reorder under the target stockpile:

$$I_i < S_i \tag{15}$$

Resupply Rate and Orders

- An order O is a quadruple $O = (i, q, \tau, f)$, where i is the identity of the product being ordered, q is the desired quantity, τ is the desired lead time, and f is an identifier of the customer.
- Let $\mathcal{O}_i$ be the set of all outstanding orders for good i placed by firm f. Define the *resupply rate* to be

$$r_i := \sum_{O \in \mathcal{O}_i} \frac{O[q]}{O[\tau]} \tag{16}$$

- This number is the rate at which i enters the inventory (if orders were not discrete.) If r_i is at least the depletion rate, the inventory will not fall, suggesting the another order condition:

$$r_i < D_i \tag{17}$$

or

$$R_i := D_i - \sum_{O \in \mathcal{O}_i} \frac{O[q]}{O[\tau]} > 0 \tag{18}$$

where R_i is the *resupply deficit*.

Order Sizing

- Firms aim to make up the resupply deficit with its next order:

$$\frac{q}{\tau} = R_i \tag{19}$$

- Presently, we simplify sizing by fully exogenizing[3] lead times, assuming that desired lead times are always fixed proportion of the firm stockpile duration:

$$\tau = rS_d \tag{20}$$

  for some constant parameter r, currently set to 0.25. Quantities are then uniquely determined to attain $r_i \geq D_i$:

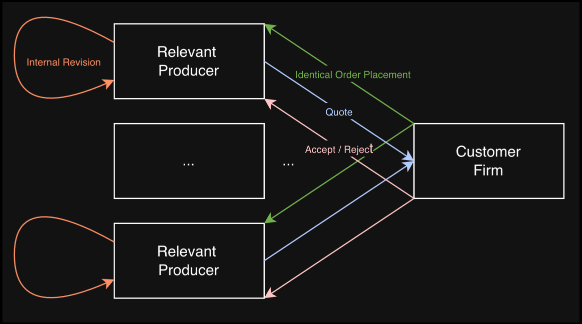
$$q = R_i r S_d \tag{21}$$

-

[3] Eventually, we intend to endogenize these desired lead times, but doing so stably requires additional modeling of the negotiation between the producer and customer over both of these quantities, which we have not done.

Order Placement

- When the order is created, the customer firm f sends it out to all suppliers of the desired good.
- Suppliers *draft plans*, which they store internally. These may not fully meet order requirements due to raw material and labor constraints, so suppliers send a *revised order* (like a quote) back to the customer.
- In our current strategy, the customer maximizes the reorder's potential contribution to the resupply rate.
- The customer sends signals to each producer whether their quote was accepted (production is pursued) or rejected (the internal plan is thrown out).



Plans: Basic Theory

- A *plan* represents an intention to get work done.
  - A producer plans to fill an order from a distributor or another producer.
  - A distributor plans to distribute goods to consumers.
- A plan puts together
  - A product
  - A quantity
  - A budget:
    - Machinery (fixed capital) costs — determined by standard techniques and current prices.
    - Input or raw material (fluid capital) costs — likewise.
    - Expected living labor costs — estimated from previous experience.
- A *debt*, equal to the budget. When the product is made and delivered, the recipient pays for it in labor hours, which offset the debt.
- Repayment of debt serves to monitor how well production is correlated with people's needs.

Plan Drafting: Order Revision

- To summarize, we have a customer firm f who has broadcast an order $O = (i, q, \tau, f)$ to a valid producer. This supplier must restrict order quantity to the maximum producible amount by its inventory:

$$q \mapsto \min\left\{q, \frac{I_g}{a_{ig}} \;:\; g \in \mathcal{G}\right\} \tag{22}$$

- The necessary machines also give an *order rejection* condition:

$$\bigwedge_{m \in M_i} I_m > \frac{\tau}{\mathcal{L}_m} \tag{23}$$

where M_i is the set of machines used to produce i, and $\mathcal{L}_m$ is the lifetime of m.

Plan Drafting: Worker Prediction

- Now the drafting firm must assign workers to the plan, first predicting the number of workers needed.
- Firms maintain their own local estimates of the per-unit living labor required to produce products in their catalog. Call this number $\tilde{l}_i$.[4]
- For a given predicted number of workers $\tilde{N}_W$, the predicted lead time is then

$$\tilde{\tau} = \frac{q\tilde{l}_i}{\tilde{N}_W \omega} < \tau \tag{24}$$

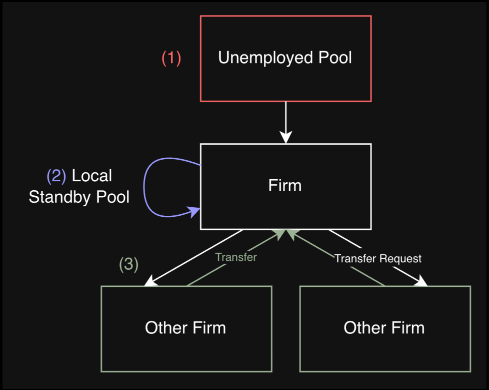
which we want to restrict to be less than the requested lead time. The working week proportion, ω, acts as a conversion between labor time and real elapsed time.

$$\tilde{N}_W = \left\lceil \frac{q\tilde{l}_i}{\tau\omega} \right\rceil \tag{25}$$

[4]This will differ between firms, but the index is omitted.

Plan Drafting: Worker Assignment

- There are three potential pools for worker assignment:
  1. The unemployed population
  2. The firm's own *standby workers*, i.e. those who are not currently working on any plan.
  3. Workers employed by other firms but available for transfer.
- Workers are assigned one at a time by the above precedence, until N_W many workers are assigned or all potential pools have been exhausted.
- In the case of (2), workers are sorted by ability, prioritizing skill. No such sorting is currently conducted for pool (1).
- Transfers of workers between firms will be discussed separately in its own section. The final number of workers assigned is $N_W \leq \tilde{N}_W$.



Plan Drafting: Final Adjustments

- Now that $N_W \leq \tilde{N}_W$ is fixed, the firm re-evaluates its *quantity* to meet the deadline:

$$\frac{q\tilde{l}_i}{N_W \omega} < \tau \tag{26}$$

$$q \mapsto \min\left(q, \frac{\tau N_W \omega}{\tilde{l}_i}\right) \tag{27}$$

- After this, the predicted lead time is updated and set back as a return order to reflect these final adjustments:

$$\tilde{\tau} \mapsto \frac{q\tilde{l}_i}{N_w \omega} \tag{28}$$

Plan Drafting: Initializing the Plan Budget

- Some budgetary and accounting information is initialized for the plan:
  - The plan labor budget:

$$L := q\tilde{l}_i \tag{29}$$

  - The machinery budget: Let p_m denote the current unit price of machine m. Then

$$M := \sum_{m \in M_i} p_m \frac{\tilde{\tau}\omega}{\mathcal{L}_m} \tag{30}$$

  - The raw materials budget: If $\mathbf{a}_i$ denotes the i^{th} row of A_{gg}, and $\mathbf{p}_g$ is the vector of current unit prices for each regular good, then this is

$$R := q(\mathbf{a}_i \cdot \mathbf{p}_g) \tag{31}$$

Plan Drafting: Social Debt

- An overall *social debt* field is then initialized:

$$V = L + R + M \tag{32}$$

- This represents the amount of labor value which the firm should be generating for society through the execution of the plan.

Plan Cycle

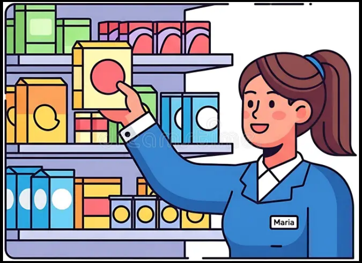
- When a firm starts executing a plan, it adds the plan budget (equal to the debt) to its account.
- The firm draws from the account to reorder inputs and machines as needed and to pay workers as the plan progresses.
- When the plan is completed, the debt is deducted from its account.
- Revenue from the sold product is added to the account.
- This is standard accounting practice for the GIC, and makes the firm transparently and publicly accountable.

Plan Drafting: Initializing Ledgers

- Finally, some 'record' fields are initialized, which will be filled in if and as a pursued plan progressed:
  - A plan inventory vector $\mathbf{I}$. This will be populated with the relevant numbers when a plan is pursued, and drain as inputs are used.
  - A plan *outlays* vector $\mathbf{U} \mapsto \mathbf{I}$, which serves as an in-kind record of inputs actually used in production, to be reported to the price controller.
  - A plan quantity remaining: $q_r \mapsto q$. Will be deducted from as the plan progresses.
  - A labor hours used scalar: $L' \mapsto 0$.
- Overall then, a *plan* P is the data structure consisting of 14 fields (here, W is the set of workers assigned to the plan):

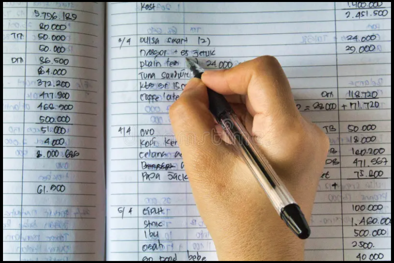
$$P = (q, q_r, \tau, i, f, g, W, L, L', M, R, V, \mathbf{I}, \mathbf{U}) \quad (33)$$

The Chosen Firm Starts the Plan



Pull inputs from inventory, reserve machine time.

Assign workers from standby, other firms (transfers), or unemployed.



Note outlays, to be revised as plan progresses.

Mark the plan as **in progress**.

Plan Progress: Simulating Production

- On each time step, if it occurs within working hours, each plan of each firm advances a step.
- If the amount produced so far has reached the original order quantity, the plan is complete.
- Otherwise, the amount produced depends on the condition of each worker:
  - The worker's skills.
  - The worker's health condition: well or ill, injured, etc.
- This amount is, in effect, the actual labor performed, l_i, as opposed to the estimate used for planning, $\tilde{l}_i$.

Plan Progress: Simulating Production

- Let $w \in W$ be a worker assigned to the plan, $\mathbf{s}_w$ the vector of aptitude levels for each relevant ability of that worker, and $\bar{s}_w$ the average over the values of $\mathbf{s}_w$. Let $\rho_w = 1$ if the worker w is healthy, and 0.5 if the worker is unhealthy. Then, the effective labor that this worker will perform during the hour is $\rho_w \bar{s}_w$.
- Then the labor performed *towards progressing the plan* during the current hour is:

$$\Delta L = \sum_{w \in W} \rho_w \bar{s}_w \tag{34}$$

- From this, an *expected* quantity produced during the hour can be computed:

$$\Delta q_e = \frac{\Delta L}{l_i} \tag{35}$$

- In the plan's last hour, we avoid overproduction: the actual quantity produced is the minimum of the above value with the quantity remaining:

$$\Delta q = \min\left(\Delta q_e, q_r\right) \tag{36}$$

Plan Progress: Simulating Production

- If $\Delta q = 0$, the plan was completed in the last step. Otherwise...
- The required inputs needed in this step are computed. If $\mathbf{a}_i$ is the ith column of A_{gg}, then the inputs needed and used are

$$\mathbf{n}_g = \Delta q \mathbf{a}_i \tag{37}$$

- *If* the full hour is worked, then the portion of a machine m with lifetime $\mathcal{L}_m$ used up during that hour is $\frac{1}{\mathcal{L}_m}$. However, it is possible that the plan finishes sometime during the final hour, since time steps are discrete but lead times are continuous. Define the portion of an hour worked as:

$$P_H = \frac{\Delta q_i}{\Delta q} \tag{38}$$

- For example, if a plan halts after producing just half of the expected quantity, then we assume it halts after 30 minutes, and $P_H = 0.5$.

Progress: Simulating Production

- For machine m with lifetime $\mathcal{L}_m$, a time step uses up $\frac{P_H}{\mathcal{L}_m}$ of its usable lifetime.
- Although the firm reserved enough machine capacity at plan start for the *estimated* plan duration, workers could wind up taking longer than expected because of illness, etc., so a machine could break down during plan execution.
- In that case, we put in an emergency order for the machine and keep trying on subsequent time steps to resume work.

Plan Progress: Simulating Production

- Recall that $\mathbf{n}_g$ is the vector of quantities of inputs needed in one time step of producing q of product i. Let $\mathbf{n}_m$ be the correponding vector of portions of machine lifetimes used up in a time step. Note that both vectors have as many rows as $\mathbf{A}$ — with 0s in rows not pertaining to the inputs or machines, respectively.
- So, barring a machine breakdown emergency, the firm updates the plan inventory on every time step: $\mathbf{I} \mapsto \mathbf{I} - (\mathbf{n}_g + \mathbf{n}_m)$.
- The firm updates labor hours used thus far: $L' \mapsto L' + P_H|W|$.
- And it updates the quantity remaining: $q_r \mapsto q_r - \Delta q$.
- Finally, each worker receives P_H labor credits. (Note that this quantity will be taxed according to the FIC. More on this soon.) These credits are also deducted from the firm's account.

Ending a Plan

- As mentioned, plans are completed when $q_r = 0$. The ending procedure is as follows:
  1. The plan is marked as **finished**.
  2. The completed order is shipped to the customer (instantly — we do not yet model shipping time or costs).
  3. The customer pays the firm in labor-time credits, which the firm counts against the plan's debt V. The firm then subracts V from its account. (This is equivalent to subtracting the whole debt from the account and then adding back the revenue from the shipped order.)
  4. Any unused inventory is subtracted from the plan *outlays*: $\mathbf{U} \mapsto \mathbf{U} - \mathbf{I}$, as well as returned to the firm's general inventory.
  5. The firm updates its local estimate of the living labor per unit of product i:

$$\tilde{l}_i \mapsto \frac{L'}{q} \tag{39}$$

  6. The firm sends the plan to the price controller, which updates the price of good i based on the plan's data, to which we turn next.

Updating Prices

- The price controller is an algorithm for updating the prices according to recently completed plans. The updating procedure is remarkably simple, and works as follows.
- Firstly, the price controller maintains estimates of all socially average per-unit *living* labor values. Essentially, it attempts to do the same thing that the firms do with their $\tilde{l}_i$ numbers, but estimated for society as a whole. Call these numbers l_i^* for each i.
- Additionally, the price controller maintains estimates of all per-unit *dead* labor values, i.e. the number of hours of indirect labor required per unit of each good. Let us call these numbers r_i^* for each i.

Updating Prices

- In order to update these numbers, the price controller maintains a queue Q containing all completed plans for all orders within a fixed time window. This means that when a new plan is completed, it is immediately added to the queue, while all plans whose timestamps fall outside of this window are dropped from it.
- Recall that every completed plan P includes an outlays vector of materials used $\mathbf{U}_P$, as well as a field L'_P denoting the number of hours of labor actually used. At the time when an order for product i is completed, the price controller uses these to update l_i^* according to:

$$l_i^* \mapsto \frac{\sum_{P \in Q} L'_P}{\sum_{P \in Q} q_P} \tag{40}$$

- i.e. it is simply the ratio of total labor expended on product i divided by the total quantity of i produced in that window.

Updating Prices

- Similarly, if $\mathbf{p}$ denotes the current vector of unit prices, then r_i^* is updated according to

$$r_i^* \mapsto \frac{\sum_{P \in Q} \mathbf{p} \cdot \mathbf{U}_P}{\sum_{P \in Q} q_P} \tag{41}$$

- I.e. it is simply the ratio of the total price of all products used in the production of i divided by the total quantity of i produced.
- Finally, the price of product i is updated to

$$p_i \mapsto l_i^* + r_i^* \tag{42}$$

- That's it! The algorithm operates in time linear in the rate of order completion, as well as the number of products in society.
- However, in our testing, the algorithm has proven to be remarkably capable of approximating labor values, snapping immediately to them from any initial condition.

Additional Notes on the Price Controller: Distributed Trust

- One of the primary but under-rated advantages of a labor time economy is the honesty and straight-forwardness of the price system. Residents of the LTE can look at the prices on goods and immediately understand what is being communicated to them by that number.
- However, this advantage can only be realized if people *trust* that the numbers are being computed correctly and without deceit.
- Open source is necessary but insufficient for this. To maximize trust, we believe that the price controller is best conceived of as a distributed ledger technology (e.g. blockchain), in which many volunteers run the same algorithm simultaneously, and in which consensus from all running instances is required for the final update to apply.

Additional Notes on the Price Controller: In-Kind Accounting

- Note that the price controller makes use of the actual inputs used in the form of the outlays vector $\mathbf{U}$. A computationally and bureaucratically cheaper alternative would be to have firms simply report the *total cost* of the inputs. In fact, these are already derivable from existing plan fields: $M + R$.
- One could then approximate r^* the same way as above, just using the $M_P + R_P$ numbers in the numerator instead of the $\mathbf{p} \cdot \mathbf{U}_P$ numbers.
- It has been witnessed that this cheaper method *does* produces convergence towards labor values (see here for details). However, the convergence rate is very slow compared to the in-kind method. Thus exclusive reliance on these proxy numbers is a bad idea.
- However, this does demonstrate a certain flexibiltiy regarding the price controller. If a firm doesn't provide meticulous accounting details of all inputs used, the above method can be employed in place of that, within the same updating procedure. However, it is important that firms provide detailed in-kind accounting whenever possible, especially in times of technological upheaval.

Additional Notes on the Price Controller: Machine Reporting

- The price controller algorithm is noteworthy in that it requires no special data other than what firms would already find themselves keeping track of. The only apparent exception to this is machines, since it seems here as if firms are meticulously keeping track of and submitting exact portions of a machine's lifespan which are used up by a plan.
- However, this could easily be replaced by some basic data collection on average machine lifespans which is provided to the price controller exogenously.
- From this, the price controller can approximate the machine use by simply taking the labor hours performed L' and dividing by the corresponding machine lifetime.
- In the case of machines which deteriorate passively over time rather than through use, such as buildings, one can simply perform the unit conversion $\frac{L'}{\omega}$ to derive the real time which elapsed over the course of the plan, and then do the same thing.

Labor Allocation

- The final pillar which needs to be surveyed within our model is our system for employment transfer.
- Any mode of production must provide a system which can reallocate labor where necessary when necessary. However, labor reassignment is an extremely disruptive process under any circumstances.
- While the disruptive nature of reassignment can not truly be avoided, we believe that the coldness and cruelty characterizing such systems can be, through a combination of cultural framing, social security, and, most importantly, the alignment of material interests between persons, firms, and society as a whole. This is therefore a sensitive and critical aspect of the system design, and we took it very seriously.
- In this department, the GIC were in hindsight very wise to insist that workers are only paid labor credits for time worked on a plan. Why is this?

Why Only Pay Workers for Time on a Plan?

- Two reasons:
  1. If workers were paid for all time considered part of the working day, then they would receive more labor credits than the total value of goods, leading to risk of overconsumption.
  2. It is crucial to aligning the interests of people with those of the broader society in terms of the ability for the division of labor to approach equilibrium conditions.
- If the economy requires more of profession A and less of profession B, then firms providing A will receive more orders, while firms providing B will receive less. Workers employed at these latter firms will find themselves less busy, and begin seeing their accounts dwindle over time.
- This provides a material incentive for the workers to *electively* transfer from one firm to another.

The Transfer System in Concept

- This is to be contrasted with capitalism, where workers are coldly dismissed by their firms, sometimes after being treated like 'family' for many years. In an LTE, this familial veneer can potentially be transformed into something genuine.
- Since workers themselves are incentivized to transfer, the economic need for firms to dismiss their workers is all but eliminated. This opens the door to all kinds of cultural framings which authentically embrace the potential for firms to provide social inclusion and security to their employees.
- For example, an LTE could foster a culture of temporary 'solidarity sabbaticals', in which workers always have a 'family' firm to fall back to, but are willing to go where they are needed from time to time.
- Alternatively, workers might embrace periodic work between several firms over the course of several years.
- There are many ways to culturally present a system of labor mobility.

The Transfer System Implementation: Busyness

- Despite the system conceptually being an elective, person driven process, our system currently handles employment transfers at the firm level. It is our hope, and assumption, that the alignment of interests between parties here makes it irrelevant where the system is actually handled in code.
- In the near future we are planning on having the algorithm more closely follow the conceptual framing.
- Rather than tracking accounts directly, persons and firms monitor their *busyness* over time, denoted b. A worker's busyness is simply the number of hours of work (on plans) that the worker receives per hour of real time.
- Worker and firm busyness is monitored and maintained through an identical exponential smoothing technique to what we previously described for demand.

The Transfer System Implementation: Busyness

- Employment transfers are enabled when a firm's overall busyness, relative to the social average, falls below a certain threshold T_B, which currently defaults to 5%. I.e. when $\frac{b-\bar{b}}{b_s} > T_B$
- If transfers are enabled, a maximum number of workers available for transfer M_W is computed as

$$M_W = \left(1 - \frac{b_f}{b_s}\right) W_f \tag{43}$$

where b_f and b_s are the busyness values for the firm and society, respectively, and W_f is the number of workers employed with the firm.
- Thus if the firm is 90% as busy as society writ large, then a maximum of 10% of its workers are available for transfer.

The Transfer System Implementation

- Transfers can happen in two separate places.

(1) During the drafting process. Specifically when trying to assign workers to a draft plan, firms who cannot find the required number of workers in the unemployed pool or their own standby pool will send requests to other firms in search of the remainder. Of course, transfers are only finalized when a plan is pursued.

(2) Once per week, firms manually check to see if their busyness is lax enough to warrant transfers. If so, then the calculated number of workers is transferred to the busiest firm in society.

- Without (2), one could hardly call our system acceptable as a stand-in for a worker-driven elective system, since at that point workers would only be allowed to leave when another firm explicitly needed extra help.
- There is another, more interesting reason why it proves necessary, but we will save this discussion for the results section.

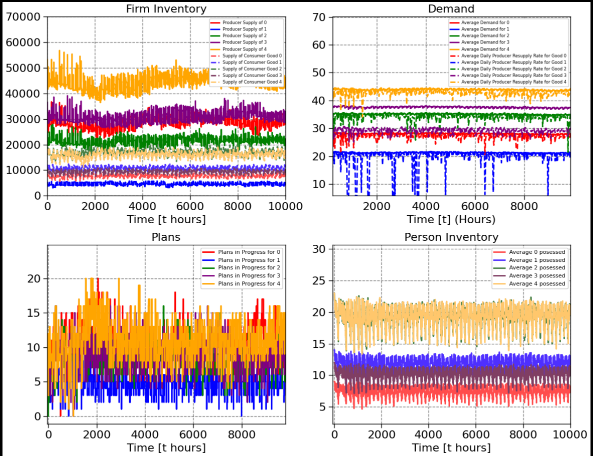
3. Results

Viewing Results in Overseer

- Our simulation has an interactive software layer in which users can run the simulation with various parameter settings and view a wide variety of aggregated data in real time.
- This interactive layer is facilitated via another open source project of Dr. Creiner's called Overseer, which allows one to easily create Desmos-style interfaces for whatever kind of model they are working on.
- I recently made a presentation on Overseer for my youtube channel, which you can find here.
- However, if you just want to jump into the LTE simulation, Overseer is included as a submodule within the Labor Time Economy Simulation GitHub repo, along with setup scripts which can be run to immediately get everything set up.
- To get started, simply visit the LTE repo and follow the instructions listed under the "Building and Running Through Overseer" section.

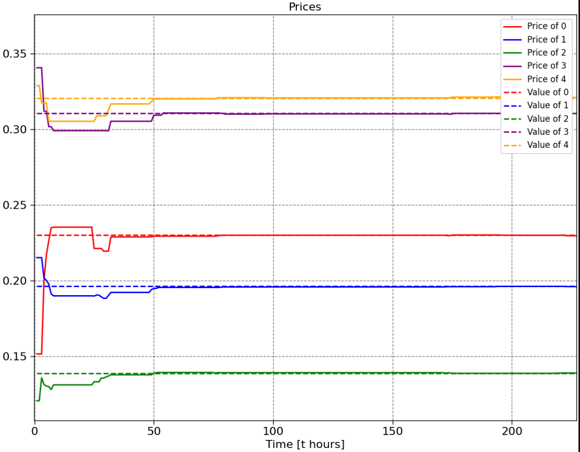
Results: Inventory and Demand

- From our testing, we can see that the drafting and reordering rules described do seem extremely effective at driving economic production. Inventories, both of firms as well as persons, appear stable.
- Economic homeostasis is observed even at extremely harsh difficulties of production which are extremely close to 1 (the logical maximum).
- Though we have not yet implemented shifting ‘tastes’ to test this, we have no reason to doubt that the system would be observed to dynamically adjust itself competently as those tasts shift.

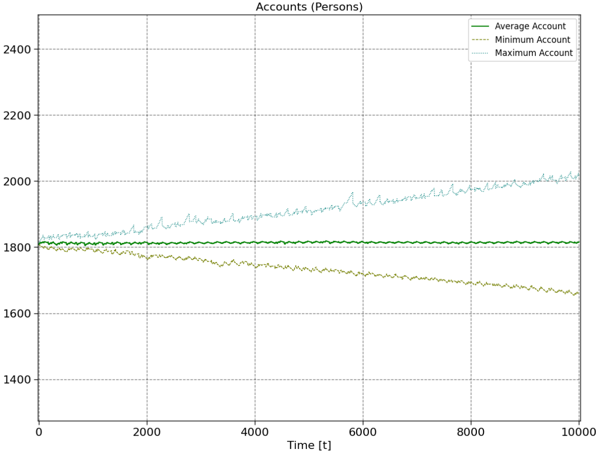


Prices

- Prices appear to stably and accurately remain approximately approximate labor values. Moreover, when initialized from equilibrium prices, we see them *immediately* snap to their labor values. This is consistent with our theoretical formal results which are available upon request.

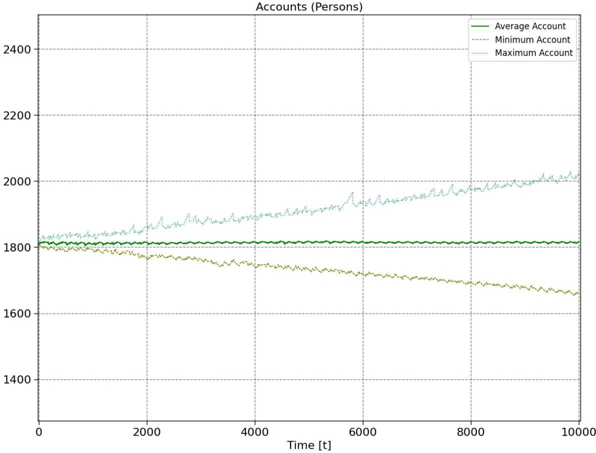


- Prices do not perfectly match the theoretical labor values predicted by the Leontief system, but this is to be expected given the modifying elements of health and job proficiency.



Prices

- The prices approxmated by the price controller do seem accurate enough to stabilize incomes over time - workers find themselves able at all times to consume naturally using their income.
- Additionally, prices appear stable and accurate enough to continue ensuring income meets demand even after income taxation through the FIC (more on this later).



Results: Prices

- Prices appear to stably and accurately remain approximately approximate labor values. Moreover, when initialized from equilibrium prices, we see them *immediately* snap to their labor values. This is consistent with our theoretical formal results which are available upon request.
- In conjunction with the labor-transfer system's ability to assert busyness equalization, the prices approxmated by the price controller do seem accurate enough to stabilize incomes over time - workers find themselves able at all times to consume naturally using their income.
- Prices do not perfectly match the theoretical labor values predicted by the Leontief system, but this is to be expected given the modifying elements of health and job proficiency.
- Additionally, prices appear stable and accurate enough to continue ensuring income meets demand even after income taxation through the FIC (more on this later).

Results: Activity Levels

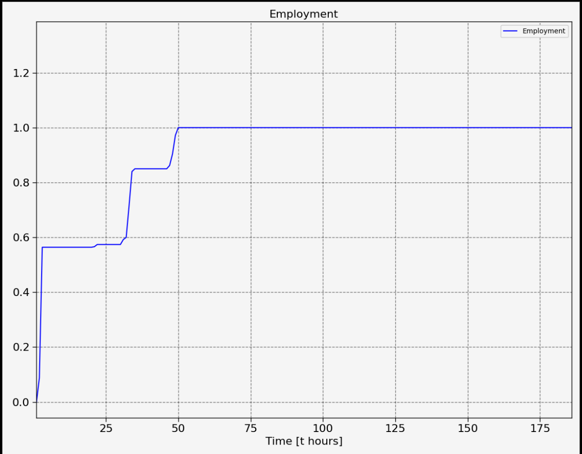
- Recall that given our fixed consumption frequencies, we have a predictable net (hourly) demand bundle $\mathbf{y} = n_p \mathbf{c}$. We can find the gross bundle by taking the Leontief inverse:

$$\mathbf{x} = n_p (I - A)^{-1} \mathbf{c} \tag{44}$$

- Due to the identity between activity levels and output in the classical Leontief framework, we can choose to also see $\mathbf{x}$ as the vector of *minimal hourly activity levels* for each sector.
- On the other hand, we can define the average activity level at a given moment of the simulation as the total quantity entering into production across all plans divided by the time span.
- In Overseer, we see long-run activity levels which converge towards these minimal levels from above, exactly as desired. Additionally, the average weekly activity levels witness the consistency of this behavior at a given moment.

Results: Employment

- Implicitly, our society has an imperitive of maintaining full employment. This is reflected in the priority given to unemployed persons when drafting plans, as well as the complete omission of any rules for ‘releasing’ workers back into the unemployed population once hired. Thus, it is no surprise that we see employment immediately reach 100% and remain there.



Results: Minimal Employment Bounds

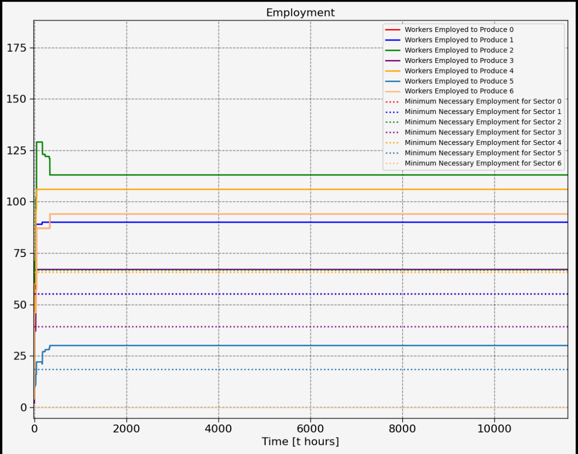
- Similarly to how $\mathbf{x}$ was used to derive minimal activity levels, it can also be used to derive minimal employment conditions.
- The entry-wise product of $\mathbf{x}$ with the living labor vector $\mathbf{l}$ reveals the sectoral labor requirements for each product. That is to say, for each product i, $l_i x_i$ represents the number of hours of labor which are needed each hour in the production of i needed to sustain demand.
- Finally, dividing this quantity by the number of hours in the working week gives us the number of hours of such hour needed per hour of the working week. Thus

$$E_i = \left\lceil \frac{l_i x_i}{TW} \right\rceil \tag{45}$$

is the *minimal* number of workers who must be consistently employed in the production of i in order to sustain social reproduction.

Results: Minimal Employment Bounds

- These lower bounds give us an objective metric by which to judge whether or not our transfer system is producting the socially necessary distribution of labor. Indeed, this is precisely what we see.
- Moreover, we see relatively stable employment, with low volatility. This means that our system is effective not just in asserting a necessary division of labor, but also in keeping the disruption to worker's lives to a minimum.



Results: Busyness Upper Bounds

- Let $b = \frac{K}{\Gamma}$ represent the average busyness of an individual in society. Here K is the average number of hours worked (on a plan) each week, while $\Gamma = 24 \times 7$ is the number of hours in a week.
- Since our society has no concept of overtime, the number of hours worked can't exceed the maximum number of hours in the working week: $K \leq TW$. Dividing both sides of this by Γ gives

$$b = \frac{K}{\Gamma} \leq \frac{TW}{\Gamma} = \omega \tag{46}$$

- Thus we have that $b \leq \omega$. Thus our conversion constant ω is also an upper bound on busyness in society.

Results: Busyness Lower Bounds

- On the other hand, since this is a labor time economy, K is not only the number of hours worked per week on average, but also their weekly income in labor credits.
- Assuming production levels are minimal for meeting net demand, the total labor required of a worker each week is known - it is simply the labor value of Γ times the consumption frequency vector $\mathbf{c}$: $\Gamma(\mathbf{v} \cdot \mathbf{c})$
- Again assuming equilibrium conditions of social reproduction, we can assert that the weekly income of a worker must be at least this amount:

$$K \geq \Gamma(\mathbf{v} \cdot \mathbf{c}) \tag{47}$$

- But then dividing both sides by Γ and remembering that $\mathbf{c}$ is normalized such that $\mathbf{v} \cdot \mathbf{c} = \omega\epsilon_c$, where $\epsilon_c < 1$ is our consumption demand constant, we have

$$b = \frac{K}{\Gamma} \geq \omega\epsilon_c \tag{48}$$

Results: The Busyness Window

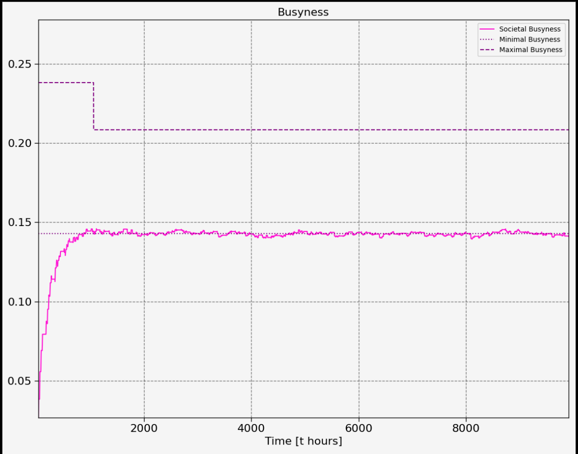
- Thus we have derived bounds on the average busyness, within which social reproduction is possible:

$$\omega\epsilon_c < b < \omega \tag{49}$$

- By multiplying all sides of this inequality by the number of hours in a week, we get from this bounds on the length of the working week.
- For example, suppose $\epsilon_c = 0.6$, and $TW = 40$. Then what this inequality amounts to is $24 \leq K \leq 40$. I.e. at minimum persons in society must work between 24 and 40 hours per week to enable social reproduction. This makes the busyness metric a potentially powerful tool in deciding how to reduce the length of the working day/week. More on this later.

Results: Busyness Observations

- As a busyness band, the lower bound represents a minimal level of busyness that workers must experience in order not to see their income atrophy over time.
- In Overseer, if we look at the busyness category of plots, we see that average societal busyness levels out at the minimum. This is exactly what we want - no excessive work being done and no overproduction.



Results: A Contradiction in the LTE

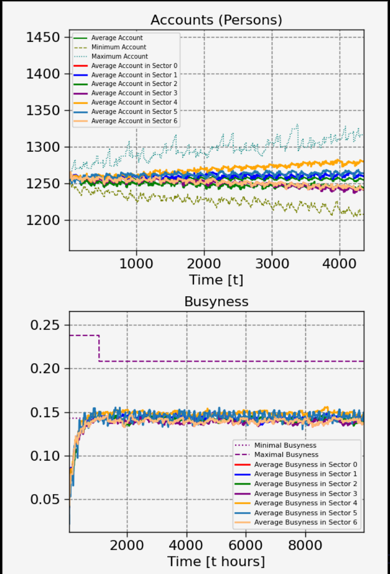
- On the one hand, we have a social imperative towards full employment. This means it is socially necessary for employment in each sector to settle *above* the actual equilibrium minimal levels.
- On the other hand, and precisely because of this, workers in a given sector will find themselves busy to a degree influenced by the ratio of workers actually employed in a sector vs the minimal number. Call this number r.
- We would expect workers to want to transfer when their sector of employment has $r < \bar{r}$. These workers *need* to transfer in order not to eventually either go broke or change their consumption habits.
- Naturally, they would want to transfer to sectors for which $r > \bar{r}$.
- For a worker in a non-busy sector to not broke, they must transfer to a busier sector. **But if this happens, workers in the busier sector will see their income deflate as a direct result of this.**

Contradiction in the LTE

- There is thus a contradiction in the *material interests of the firm and the individual.*
- Due to the full employment imperative, in an LTE workers will occasionally need to transfer to another firm *even when* that firm doesn't need the labor.
- This threatens the perceived livelihood of the workers at that firm. This may be a place where negative feedback systems are necessary in an LTE. Or it may be something which can be solved effectively within the superstructure. Regardless, it is worth taking head of this.

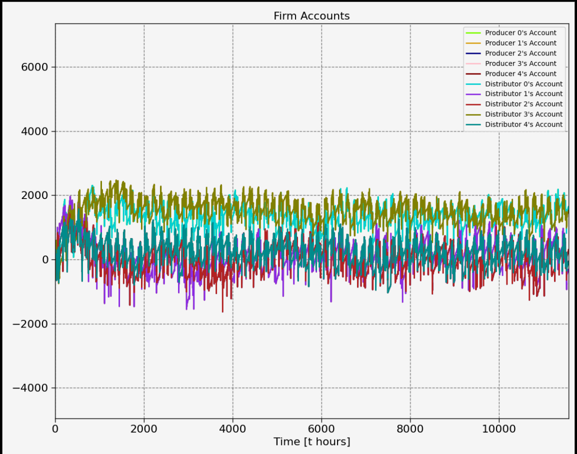
Results: Busyness Equalization

- This is also why we found it necessary to allow for workers to transfer to firms *not just* when a firm needs the help, but also when the workers themselves are not busy enough. Doing this proved sufficient for busyness, and with it accounts, to equalize over time.



Firm Accounts

- At every plan start, the firm creates its budget funds *ex nihilo* (as fiat currency), along with a corresponding debt.
- The budget is added to the firm's account.
- The firm spends the account down — asynchronously — to keep up its supplies.
- When the firm completes a plan and ships an order, it gets revenue for it.
- The revenue offsets the debt.
- Thus every firm's account should rise and fall, but hover around 0.



Taxation With the FIC

- As proof of concept of a mathematically sound system of taxation, we imagine a public sector which wishes to provide a portion of the net product 'free' to its population.
- The untaxed portion we seek to find is α, what the GIC call the FIC, or *factor of individual consumption*, given by

$$\alpha = \frac{K_p}{K} \tag{50}$$

where K is the total rate of labor, which can be spilt into contributions towards free and paid commodities:

$$K = K_f + K_p \tag{51}$$

Current Price Approximation

- To obtain α, we use known free and paid portions of the net product $\mathbf{n}$.

$$\mathbf{n} = \mathbf{n}_f + \mathbf{n}_p \tag{52}$$

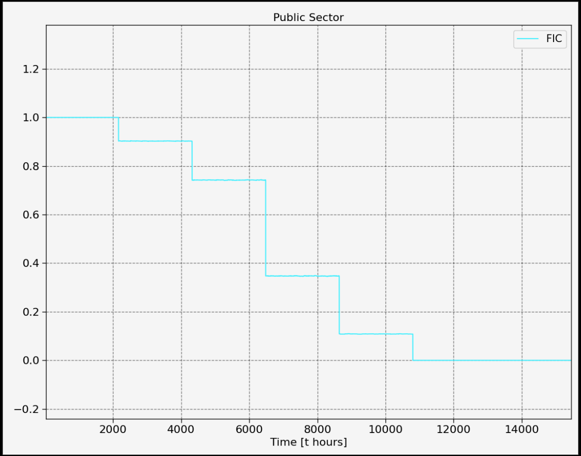
- Assuming an accurate price vector $\mathbf{v}$, we can find α directly as a function of $\mathbf{n}_p$.

$$\alpha = \frac{K_p}{K} \approx \frac{\mathbf{v} \cdot \mathbf{n_p}}{\mathbf{v} \cdot \mathbf{n}} \tag{53}$$

- It can be shown that this formula equals the one given by the GIC itself under the assumptions of our model. The proof can be found here.

Taxation With the FIC

- With the free goods checkbox clicked, a new random good will be made free every Δ_F hours.
- As this happens, the FIC will adjust downward from 1. Workers will receive shrinking wages, but as this happens their need for wages will also decrease, with more and more goods being provided as free public goods.

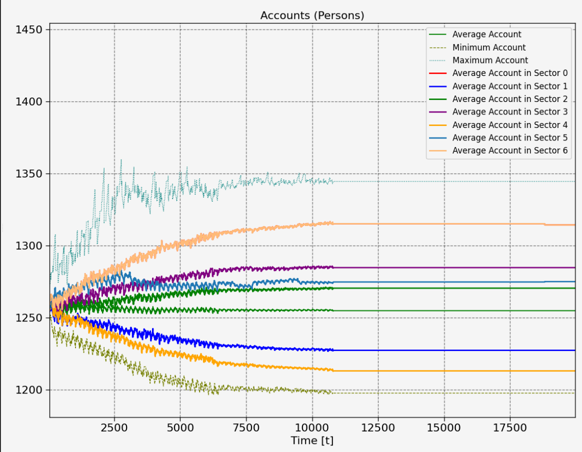


Results: The Complete Abolition of Money

- We are thus able to witness a toy transition to 'full communism'.
- While such a process could conceivably be witnessed in any economy that uses an income tax as its sole source of revenue, we have therefore witnessed labor credits as a viable mechanism by which money can, one day, be fully abolished, without any necessary economic disruption.

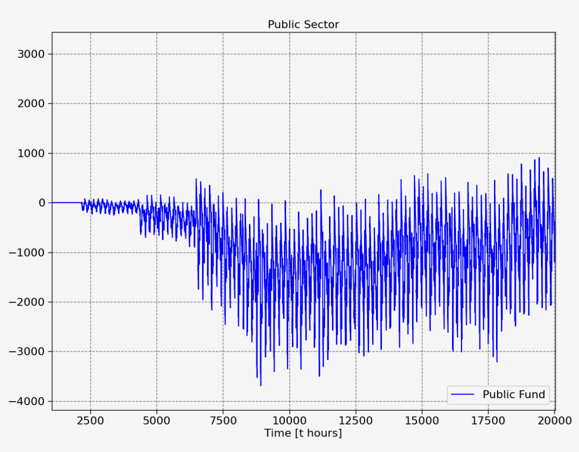
Results: The Complete Abolition of Money

- Indeed, in this plot of roughly two years, where goods are made free every three months, after just after year 1, accounts 'flatline', reflecting the fact that labor credits are no longer in use in the economy.
- This does not destabilize the simulation, because in our model the decision to transfer is based on busyness as a proxy for the rate of change in their account.
- However, when a person no longer relies on labor credits for their consumption, their incentive to transfer their employment disappears. This deserves further consideration.



Public Sector Fund

- With taxation, we get the notion of public revenue and expenditure. What do the state coffers look like over time?
- The "public fund" is added to at the point of FIC taxation, and deducted from when free goods are 'purchased'. From our limited testing, the public fund appears to be bounded, but its volatility increases as more goods become free.
- Because the computation of the FIC is only as accurate as the price controller is at approximating true labor values, this method does not guarantee zero public account drift. Future implementation of public funded retirement may fare better with an algorithm *aiming to balance the public account*.



Adjustments of the Working Week Length

- The length of the working week should only be as long as necessary for reproduction, maximizing free time during plans, while also keeping incomes consistent. The level of utilization of the working week can be quantified by

$$\epsilon_c = \frac{b_c}{\omega} \tag{54}$$

- Since b_c is not public information, we estimate:

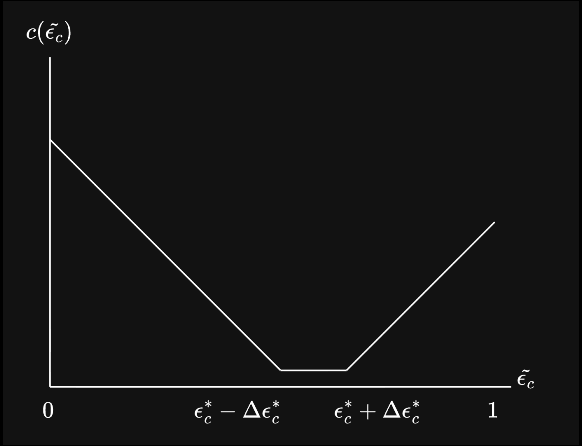
$$\tilde{\epsilon_c} = \frac{1}{n_p} \sum_{P \in \mathcal{P}} \frac{P[b]}{\omega} \tag{55}$$

and modulate it with an exponent γ, giving an upwards correction when there is variance in the distribution of $P[b]$. The assumption is the variance in busyness necessary for stability.

$$\tilde{\epsilon_c} = \left(\frac{1}{n_p} \sum_{P \in \mathcal{P}} \left(\frac{P[b]}{\omega} \right)^{\gamma} \right)^{1/\gamma} \tag{56}$$

Adjustments of the Working Week Length

- Once we've estimated ϵ_c, we set a target working week utilization ϵ_c^* (with some radius $\Delta\epsilon_c^*$). ω is updated in the direction that fulfills $\epsilon_c^* - \Delta\epsilon_c^* < \tilde{\epsilon_c} < \epsilon_c^* + \Delta\epsilon_c^*$.

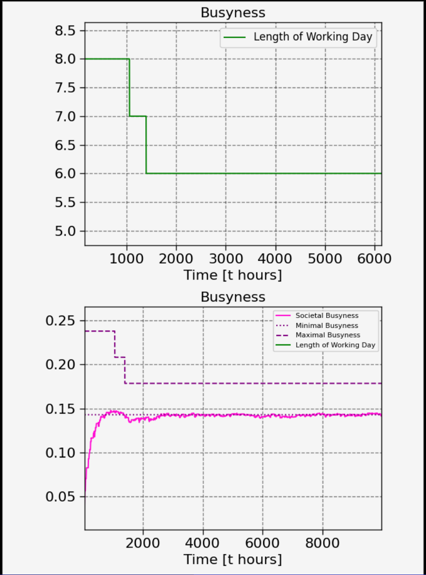


Limitations of ϵ_c^*

- Although theoretically optimal, we cannot set $\epsilon_c^* = 1$. When $\omega = b_c$, the upper bound coincides with the lower bound for reproducibility, and the system is unable to meet such strict criteria. A system capable of this must:
  (1) be "smart" enough to enforce that every worker always be assigned to a plan producing the correct commodity that contributes to meeting the final consumer demand.
  (2) ensure every labor hour spent at a firm goes towards production, leaving no room for administrative or other tasks.
- While (1) may be feasible, (2) is not realistic. Thus, we aim to design signals and checks to inform ϵ_c^*, rather than squeeze all inefficiency out of the model.

Results: Working Week Reductions

- As it stands, $\epsilon_c^* = 0.7$ seems safe. Intuitively, the number $\frac{1}{\epsilon_c^*}$ can be thought of as the degree to which society is willing to tolerate the maximum length of the working week differing from the minimal length.
- Each week, the simulation surveys societal busyness data and adjusts the working day length up or down as necessary, eventually settling to this ratio.
- If the working week's (and with it maximal busyness) adjusts much further than this, dangerous feedback loops start happening.



4. Conclusion

Conclusion – Summary

- We model a labor-time economy as a simulated interaction among autonomous agents.
- This LTE is a *self-regulating system* following *cybernetic principles*.
- Firms draft and execute plans, and supply meets demand, in a *decentralized* and *distributed*, *self-stabilizing* system.
- This honors the spirit of the GIC's proposal — promoting self-determination and the *free association of producers*.
- Given reasonable assumptions about individuals' and firms' behavior, this stability emerges as a cumulative effect.
- We hope that this work encourages people to consider the concept of a labor-time economy as a realistic and practical approach to building a post-capitalist world that fosters freedom and human flourishing.

Future Work

- Build out the public sector —
  - Retirement.
  - Education and health care.
  - Unemployment support, retraining, and appropriate incentives.
  - Care work.
  - Public infrastructure — capital investments.
- Improve the modeling of fixed capital —
  - Investments whose value depreciates over time independently of use and whose amortized costs must be applied rationally to revenue.
  - Job training.
- Model services, where the value produced cannot be precisely tied to "time on the job."

More Future Work

- Generalize to joint production —
  - Multiple production techniques fo the same product, with different respective requirements and costs.
  - Accounting of different byproducts arising from different techniques.
- Model more complex skills requirements and labor heterogeneity.
- Model product horizons —
  - The development of new products.
  - The phasing out of unwanted products.
- Model firm lifetimes —
  - The creation of new firms.
  - The disbanding of firms in response to changing needs or conditions.

Evem More Future Work

- Model spontaneous improvements in productivity — leading to more free time, higher standards of living, or both.
- Model trade with a capitalist economy.
- Model competition based on quality.

Open Questions

- Can the various agents' interests be aligned so as to ensure that productivity-improving innovations propagate from firm to firm?
- Can we integrate in-kind accounting into the system to control the use of scarce resources and to restrict harmful side effects such as pollution and environmental degradation?
- How can higher-level (System V) decisions and regulations be applied to the autonomous elements of the LTE?
- You (yes, you!) if you have a request, you can open an issue, if you want to contribute, you can open a PR.

Join Us!

- Find us on GitHub at https://github.com/BC-LTEWG/Labor-Time-Economy-Simulation
- Become a contributor — write an issue, submit a PR.

Our Team[5]

Alexander Creiner

Devin Lim

Nathan Lim

Amittai Aviram

[5] Many thanks to Boston College for support through its Undergraduate Research Fellowship program.

Our Team

Ivan Nicholas Williams

Joshua Liu

Salamun Nuhin

Zhejun Zhang